

PROJECT : 14

Object : Anti-Theft Flooring System using Raspberry Pi

A comprehensive B.Tech IoT laboratory project designed to demonstrate practical implementation of embedded systems and IoT principles for security applications. This project integrates pressure sensing technology with microcontroller processing and IoT connectivity to create an intelligent floor-based security system.

Project Objective

The primary objective of this project is to develop an intelligent flooring system that can detect unauthorized intrusions by monitoring pressure changes on the floor surface. The system serves as an early warning mechanism for security breaches in restricted or valuable areas.

Key Goals:

- Detect unauthorized personnel movement through pressure sensing
- Process sensor data using Raspberry Pi microcontroller
- Generate immediate audible alarms using buzzer modules
- Transmit real-time notifications via IoT platform
- Provide reliable, cost-effective security solution

Problem Addressed:

Traditional security systems often require expensive infrastructure and complex installations. This project offers a simple, scalable solution that can be deployed in various environments without extensive modifications.

Components Required

The system requires both hardware and software components working in coordination.

Hardware:

- **Raspberry Pi 4**
Primary microcontroller with 4GB RAM, GPIO pins, and built-in Wi-Fi
- **Pressure Sensors**
FSR sensors (0–100N range)
- **Buzzer Module**
Active buzzer (85–95 dB alarm)

Additional Hardware:

- Power Supply (5V 3A adapter, battery optional)
- Wi-Fi
- Breadboard, jumper wires

Software Stack:

- OS: Raspberry Pi OS (64-bit)
 - Language: Python 3.x
 - Libraries: RPi , GPIO, time, requests
 - Notification: Email
 - IDE: Thonny
-

System Theory & Working Principle

The system works on **pressure detection and signal processing**.

Steps:

1. **Pressure Detection**
Sensors detect force and change resistance
2. **Signal Processing**
Raspberry Pi reads analog values via ADC
3. **Alert Generation**
Buzzer activates + notification sent

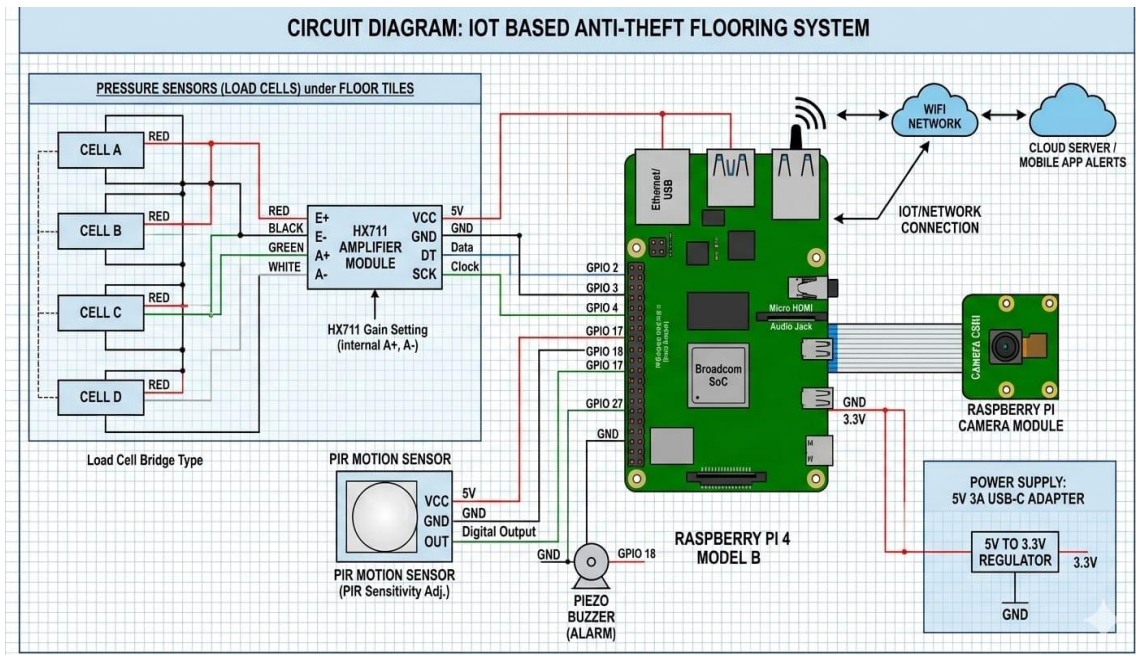
IoT Integration:

- Sends HTTP POST request to server
- Logs timestamp, pressure, location
- Sends email/SMS alerts
- Updates dashboard
- Stores historical data

Working Algorithm:

Initialize GPIO → Read sensor → Compare threshold →
If exceeded → Activate buzzer + send alert → Log data → Repeat

Circuit Diagram & Connections



GPIO Configuration:

- Pressure Sensor 1 → GPIO 17
- Pressure Sensor 2 → GPIO 18
- Pressure Sensor 3 → GPIO 19
- Buzzer → GPIO 21
- ADC (MCP3008) → SPI Interface
- Power → 5V pin

Wiring Steps:

1. Connect ADC to Raspberry Pi
2. Attach pressure sensors via voltage divider
3. Connect buzzer using transistor
4. Provide 5V power supply

⚠ Safety: Always disconnect power before wiring.

Applications & Use Cases

- Bank vaults & ATMs
- Museums & galleries
- Retail stores
- Smart homes

- Warehouses
- Data centers

Benefits:

- Prevents theft
 - Reduces guard costs
 - Improves security automation
-

 Safety Precautions**Electrical Safety:**

- Disconnect power before wiring
- Use proper fuses
- Avoid short circuits

Component Protection:

- Do not exceed 3.3V GPIO
- Use resistors
- Add protection diodes

Troubleshooting:

- Pi not booting → Check power/SD card
 - Sensor not working → Check ADC
 - Buzzer issue → Check transistor
 - No alerts → Check Wi-Fi/API
 - False alarms → Adjust threshold
-

 Conclusion & Future Enhancements**Conclusion:**

The project demonstrates integration of pressure sensing, microcontroller processing, and IoT connectivity to create an effective anti-theft system.

Achievements:

- Real-time detection
- Efficient Python processing
- IoT-based alerts

- Scalable design

Future Enhancements:

1. Machine learning detection
2. Camera integration
3. Multi-sensor fusion
4. Mobile app development
5. Wireless sensors
6. Cloud analytics

Learning Outcomes:

- Embedded systems
 - IoT architecture
 - Hardware-software integration
-